

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B. Tech. Examination (EC/ME/CL/EE)

November -2012

MA 202: Engineering Mathematics-III

Date: 26.11.2012, Monday

Time: 01:30 p.m. to 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.

SECTION-I

Q-1 (a) Find the Laplace transformation of (i) $\cos^3 2t$ (ii) $\left(t - \frac{1}{2}\right)^2 u\left(t - \frac{1}{2}\right)$. [04]

(b) Find the Fourier series for the function $f(x) = x^3$ in the interval $-\pi \leq x \leq \pi$. [03]

Q-2 (a) Solve the equation by the Laplace transforms method
 $y'' + y = \sin 3t$; $y(0) = y'(0) = 0$. [05]

(b) Solve the equation $x^3 + 6x^2 + 9x + 4 = 0$ by Cardon's method. [05]

(c) Find the half range cosine series for the function $f(x) = x^2$ in the range $0 \leq x \leq \pi$. [04]

OR

Q-2 (a) Solve the equation $x^4 - 12x^3 + 41x^2 - 18x - 72 = 0$ by Ferrari method. [05]

(b) Find the Fourier series for [05]

$$f(x) = -\pi; \quad -\pi < x < 0 \\ = x; \quad 0 < x < \pi \quad \text{and hence deduce that } 1 + \frac{1}{3^2} + \frac{1}{5^2} + \dots = \frac{\pi^2}{8}.$$

(c) Find Laplace transformation of the function (i) $3e^{-2t} \cos 2t + \frac{1}{3}t^{1/2}$ (ii) $\frac{1 - \cos 2t}{t}$ [04]

Q-3 (a) Find the Fourier series for the function
 $f(x) = 0; \quad -2 < x < -1$ [05]

$$= k; \quad -1 < x < 1$$

$$= 0; \quad 1 < x < 2$$

(b) Solve the equation by the Laplace transforms method
 $y'' - 3y' + 2y = 4e^{2x}$; $y(0) = -3, y'(0) = 5$. [05]

(c) Using Convolution theorem, obtain the value of $L^{-1}\left(\frac{1}{s^2(s-1)}\right)$. [04]

OR

Q-3 (a) Find the Fourier series of the periodic function
 $f(x) = \pi + x; \quad -\pi < x < 0$ [05]

$$= \pi - x; \quad 0 < x < \pi$$

(b) Solve the equation $x^3 - 27x + 54 = 0$ by Cardon's method. [05]

(c) Solve the equation $x^4 - 10x^3 + 35x^2 - 50x + 24 = 0$ by Ferrari method. [04]

SECTION-II

- Q-4 (a) State Stoke's theorem. [02]
- (b) The differential equation of a simple pendulum is $\frac{d^2x}{dt^2} + \omega_o^2 x = F_o \sin nt$. If initially $x=0, \frac{dx}{dt}=0$, determine the motion with the condition $\omega_o \neq n$. [05]
- Q-5 (a) A Particle moves along the curve $x=t^3+1, y=t^2, z=2t+5$, where t is the time. Find the components of its velocity and acceleration at time $t=1$ in the direction $2\hat{i}+3\hat{j}+6\hat{k}$. [05]
- (b) Show that $\vec{v} = 2xyz\vec{i} + (x^2z + 2y)\vec{j} + (x^2y)\vec{k}$ is irrotational and find the scalar function $u(x,y,z)$ such that $\vec{v} = \text{grad}(u)$. [05]
- (c) Find a unit vector normal to the surface $z^2 = x^2 + y^2$ at the point $(1,0,-1)$ [04]
- OR
- Q-5 (a) Find the directional derivative of $\text{div } \vec{u}$ at the point $(1,2,2)$ in the direction of the outer normal of the sphere $x^2 + y^2 + z^2 = 9$ for $\vec{u} = xy\vec{i} + xy^2\vec{j} + z^2\vec{k}$. [05]
- (b) Find the divergence and curl of the vector field $v(x,y,z) = (x^2 - y^2)\vec{i} + 2xy\vec{j} + (y^2 - xy)\vec{k}$ at the point $(1,0,0)$. [05]
- (c) Show that the Gradient field describing a motion is irrotational. [04]
- Q-6 (a) $\int_{(1,2)}^{(3,4)} (xy^2 + y^3)dx + (x^2y + 3xy^2)dy$ is independent of the path joining the points $(1,2)$ and $(3,4)$ and hence evaluate the integral. [05]
- (b) Using Stoke's theorem to evaluate $\int_C (x^2 + y^2)dx - 2xy dy$ where C is the rectangle bounded by $x = \pm a, y = 0 \& y = b$. [05]
- (c) Evaluate by Green's theorem $\int_C e^{-x} \sin y dx + e^{-x} \cos y dy$ where C is the rectangle with vertices $(0,0), (\pi,0), (\pi, \frac{\pi}{2}), (0, \frac{\pi}{2})$. [04]
- OR
- Q-6 (a) Using Divergence theorem to evaluate $\iint_S \vec{F} \cdot d\vec{S}$ where $\vec{F} = x^3\vec{i} + y^3\vec{j} + z^3\vec{k}$ and S is the surface of the sphere $x^2 + y^2 + z^2 = a^2$. [05]
- (b) State Green's theorem and using theorem to evaluate $\int_C (x^2 + xy)dx + (x^2 + y^2)dy$ where C is the square formed by lines $y = \pm 1, x = \pm 1$. [05]
- (c) If a force $\vec{F} = 2x^2y\vec{i} + 3xy\vec{j}$ displaces a particle in the xy -plane from $(0,0)$ to $(1,4)$ along a curve $y = 4x^2$. Find the work done. [04]
